

Layer-Time Lens on a Frozen DiT Image Editor

Instrument → Causal tests → Internal-image mechanism



Frozen (zero training)



Offline fit once, zero-shot reuse

1 WHAT WE STUDY

decode
internal states

2 HOW WE READ DEPTH × TIME

reading ≠ causality

3 WHAT WE MANIPULATE

mechanism
emerges

4 EARLY DECIDE, LATE TRANSLATE

meratation &
levers

5 WHAT IT ENABLES



noisy target tokens + clean condition (t=0)

concatenated image stream

60
DiT
blocks

post-block
residual taps
(bfloat16)

Qwen-Image-Edit
frozen end-to-end

Depth lens
intermediate $h \rightarrow$
frozen output head



Time lens
 $\hat{x}_0 = x_t - \sigma_t v$
VAE decode



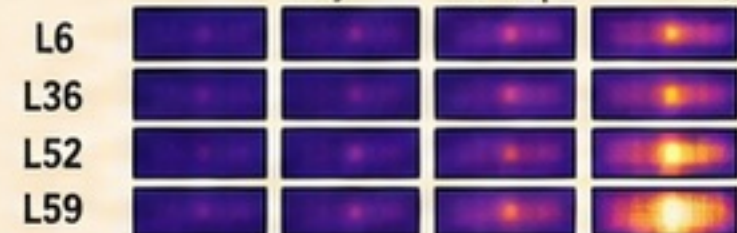
Tuned lens
ridge per (layer, step)
render mid-stack image



Differential
edit run – unchanged run
same seed $\rightarrow |\Delta|$



Layer × Time
early → late steps



CLIP readout
(crop, zero-shot)

True CFG: identical on cond & uncond (2 forwards)

Predict → Look at every image → Gate (bit-exact)

Linear probe (3-way, cross-seed)

Ablate



zero / mean
rows
stress necessity

Transplant



donor →
recipient
test sufficiency

Steer



add $\alpha \cdot d$
(no donor)
test writability

Readable?



Necessary?



Writable?



1 Injection $\leq$ L36
prompt enters



2 Decision by L6
probe $\gg$ frozen head



3 Target-image code L6–L51



4 Translation L52–L54
sharp, ~ 1 layer



5 Output L59
final red edit



anti-image if
wrong (velocity)
convention



5 WHAT IT ENABLES



Held-out early selection
($\sim 68\%$ compute)



CFG after step 2
($\sim 45\%$ forwards)



Transfers across edit families
(held-out)